

A Bayesian Model of Diabetes Risk: Causal Inference and Counterfactual Analysis Using ENSANUT 2024 Data

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Abstract

This study evaluates the impact of physical activity, sedentary time, and sleep duration on high diabetes risk among Mexican adults using ENSANUT 2024 data. The analytic sample included 1,501 participants, of whom 31.2% were classified as high risk. We applied both frequentist and Bayesian logistic regression models (via MCMC) to assess these associations. Findings revealed that older age, increased sedentary time, and fewer walking days per week correlated with higher diabetes risk. Population-level intervention simulations based on posterior samples showed that increasing walking days or reducing sedentary time by one standard deviation decreased the estimated high-risk prevalence from 31.3% to 29.7% and 29.0%, respectively. A targeted population intervention (walking at least 4 days/week and sitting for no more than 6 hours/day) resulted in an expected 1.7 percentage points reduction in absolute risk. These results suggest that increases in walking frequency and reductions in sedentary time could lower the prevalence of high diabetes risk in the Mexican adult population.

Key words: Bayesian logistic regression, causal inference, counterfactual analysis, diabetes risk

1. Introduction

Type 2 diabetes accounts for most diabetes cases worldwide. Its prevalence has increased substantially in recent decades alongside population ageing and changes in lifestyle (Darraj, 2023). Genetic factors certainly contribute to individual risk; however, a large body of evidence highlights modifiable lifestyle behaviours as key drivers of diabetes risk at the population level. These behaviours include physical activity, sedentary time, and sleep. Understanding which behaviours play a significant role is therefore critical to effective prevention strategies.

Sedentary behaviour has been shown to act as an independent risk factor for diabetes rather than simply the absence of exercise (Hamilton et al., 2014). In parallel, sleep duration has been consistently linked to diabetes risk. Both short and long sleep are associated with higher risk and indicate a U-shaped relationship (Darraj, 2023). These findings imply that lifestyle behaviours

should be modelled as distinct and potentially non-linear predictors rather than as a single dimension of activity.

From the modelling perspective, these relationships form a complex system involving multiple behavioural and demographic factors. Logistic regression provides a natural framework for modelling binary health outcomes such as high diabetes risk. A Bayesian formulation further allows uncertainty in model parameters to be quantified through posterior distributions. It also enables counterfactual simulation to assess how changes in population behaviour may affect population-level risk.

This study applies a Bayesian logistic regression model to data from Mexico’s 2024 National Health and Nutrition Survey (ENSANUT 2024). We investigate the relationship between lifestyle behaviours and high diabetes risk in an adult population. Specifically, we aim to address the following questions:

- Which lifestyle behaviors act as the primary drivers of high diabetes risk?
- What is the relative contribution and uncertainty associated with each of these risk factors?
- Under counterfactual scenarios, what is the expected shift in population-level risk if targeted behavioral changes were implemented?

2. Data description

2.1. Data Sources

The analytic dataset was constructed using a publicly available Kaggle dataset (*Diabetes_Mexico.csv*), derived from multiple modules of Mexico’s 2024 National Health and Nutrition Survey (ENSANUT 2024). This primary file contains anthropometric measures, biochemical indicators, and a clinically derived diabetes risk classification, including only participants with recorded serum glucose values to ensure a consistent clinical baseline.

To obtain lifestyle variables not fully represented in the primary file, we merged data from the ENSANUT 2024 physical activity module (*actividad_fisica_ensanut2024.csv*) using the shared participant identifier (*folio_int*). Across all lifestyle variables, survey non-response codes (88), “don’t know” codes (99), and inability codes (55) were recoded as missing. After restricting the data to complete cases on all lifestyle predictors and core covariates, the final analytic sample included 1,501 participants (61.6% female) with a mean age of 44.4 years (SD = 14.0).

2.2. Response categorization

We constructed the response as a binary indicator of high diabetes risk, derived from the *riesgo_diabetes_cat* variable. The original data classifies individuals into three categories according to American Diabetes Association (ADA) clinical criteria: (0) low or no risk, (1) moderate risk associated with overweight or obesity without glycemic alteration, and (2) high

risk, presence of glycemic alteration consistent with prediabetes criteria. For this study, participants in category 2 were coded as high risk, whereas all others were coded as low risk. Ultimately, 469 of the 1,501 participants (31.2%) were classified as high risk.

2.3. Predictors

Physical activity was assessed across three dimensions. Vigorous activity days were highly zero-inflated, as 71.2% of participants reported zero days of vigorous activity lasting at least 10 continuous minutes in a typical week. Due to this heavy skew, we recoded as a binary variable indicating any vigorous activity versus none. Moderate activity and walking were similarly measured as days per week with at least 10 continuous minutes of activity. We retained these two measures as continuous predictors. On average, participants reported 3.1 days of moderate activity per week ($SD = 3.0$) and 4.0 days of walking per week ($SD = 2.9$).

Sedentary time was calculated by converting self-reported daily sitting hours and minutes into a single continuous measure in hours per day. The sample mean was 4.3 hours per day ($SD = 3.0$). The data distribution was right-skewed and extended to nearly 16 hours per day. This pattern indicates that while most participants reported moderate sedentary time, a subset reported much higher levels.

Sleep duration was recorded as a five-level ordered categorical variable. The most frequent response was 8 hours per night, followed by 7 hours and 6 hours. The extreme categories (≤ 5 hours and ≥ 9 hours) were roughly equally represented in the sample. We designated 7 hours as the reference category because it serves as a mid-range benchmark for evaluating both shorter and longer sleep patterns. This approach is consistent with prior literature, as previous studies have reported a U-shaped association between sleep duration and diabetes risk. However, some studies have found that increased risk is mainly associated with shorter sleep duration (Darraj, 2023).

2.4. Exploratory Data Analysis

Before modelling, we first explored the bivariate associations between each predictor and high diabetes risk, as shown in Figure 1. High-risk participants are generally older than their low-risk counterparts. Regarding vigorous activity, inactive participants have a slightly higher proportion of high-risk classification (31.8%) than active participants (29.9). For walking frequency, high-risk participants tend to walk less days a week. Sedentary time displays substantial overlap between the two risk groups, with only a slightly heavier right tail in the high-risk group. Finally, sleep duration does not demonstrate a distinguishable pattern. The highest proportion of high diabetes risk occurs among participants sleeping 6 hours (33.6%), while the lowest is found among those sleeping 8 hours (29.7%).

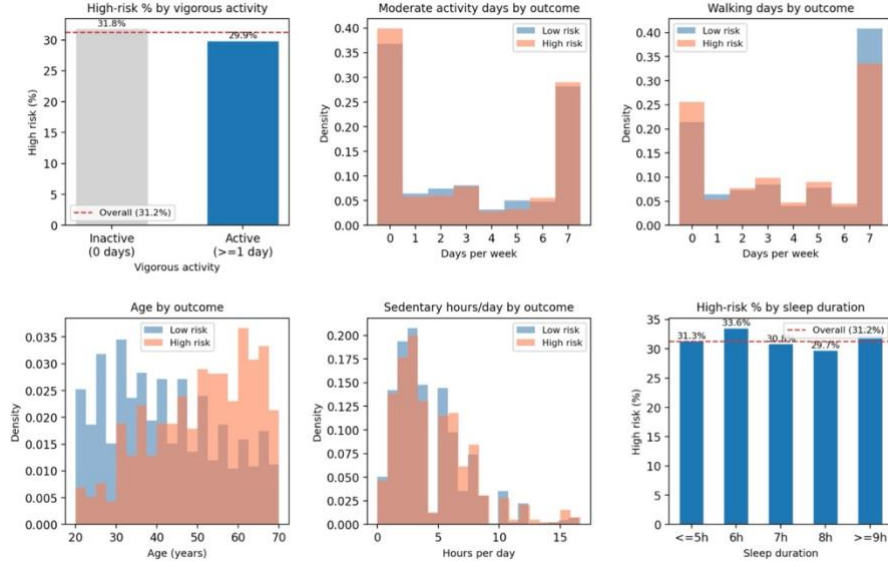


Figure 1: Bivariate associations between lifestyle predictors and high diabetes risk

3. Methodology

3.1. Causal Structure (DAG)

In the causal structure (Figure 2), age and sex are treated as baseline confounders as they are assumed to directly affect both lifestyle exposures and diabetes risk. The lifestyle variables of interest include physical activity, sedentary time, and sleep duration. These variables are assumed to have a direct effect on diabetes risk. Additionally, physical activity is assumed to influence both sleep duration and sedentary time. We also included an unobserved node (U_Diet) to represent unmeasured confounding from dietary pattern. Diet is assumed to be related to both lifestyle behaviors and diabetes risk, but it was not captured in our ENSANUT dataset.

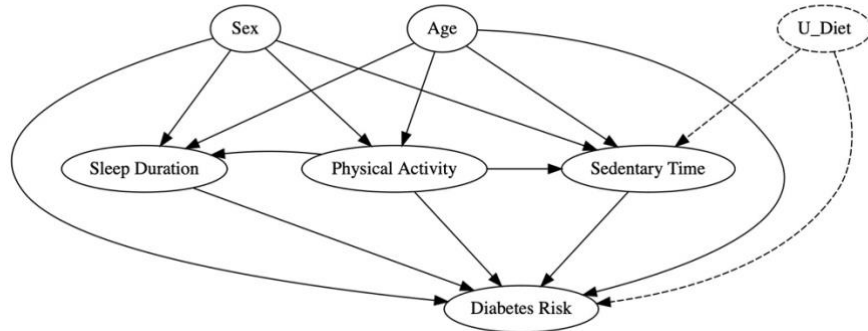


Figure 2: Directed acyclic graph specifying the assumed causal structure

3.2. Model Formulation

Let $Y_i \in \{0, 1\}$ denote the binary outcome for participant i . We set up a logit function to model the probability that participant i is classified as high diabetes risk. The model is specified as:

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \beta_0 + \beta_{\text{age}} \text{age}_i + \beta_{\text{sex}} \text{sex}_i + \beta_{\text{sed}} \text{sedentary}_i + \beta_{\text{vig}} \text{vigorous}_i + \beta_{\text{mod}} \text{moderate}_i + \beta_{\text{walk}} \text{walking}_i + \text{sleep terms}$$

Age and sedentary hours were mean-centred to improve interpretability. As a result, the intercept represents the log-odds of high diabetes risk for the reference group in each trait. The model assumes linearity in the log-odds and conditional independence of observations. As reflected in the DAG, age and sex are treated as key confounders

Prior to model fitting, the analysis was restricted to complete cases. Implausible values were also removed using predefined plausibility bounds. The model was initially fit using maximum likelihood to establish a frequentist baseline. It was subsequently evaluated within a Bayesian framework to enable full uncertainty quantification and counterfactual simulation.

3.3. Frequentist Logistic Regression

We fit a frequentist logistic regression model using maximum likelihood estimation in the Python library statsmodels. Coefficients are reported together with 95% Wald confidence intervals and odds ratios. This frequentist model serves as a reference benchmark against which the Bayesian posterior estimates are compared in the Results section.

3.4. Bayesian Logistic Regression

We placed weakly informative priors on all model parameters. The intercept was assigned a broader prior $\beta_0 \sim N(0, 1.5^2)$ to reflect greater uncertainty about the baseline log-odds. All slope coefficients were assigned $\beta_0 \sim N(0, 0.5^2)$. Prior predictive checks indicated that these priors produced a reasonable range of simulated values before observing the data. Figure 3 shows prior predictive trajectories for three key predictors, and Figure 4 shows the prior predictive distribution of simulated prevalence.

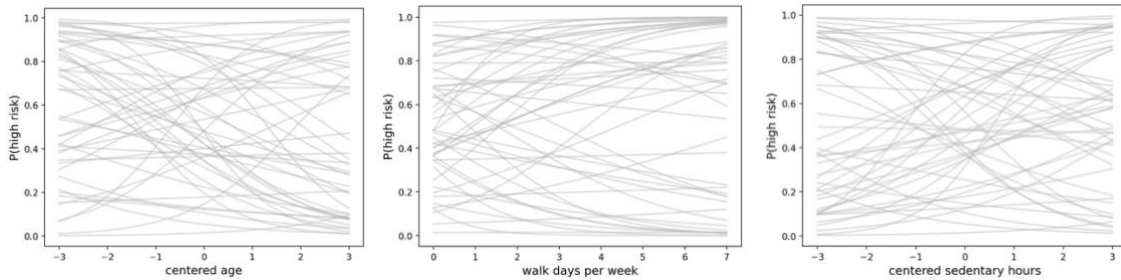


Figure 3: Prior predictive trajectories for age, sedentary time, and walking days

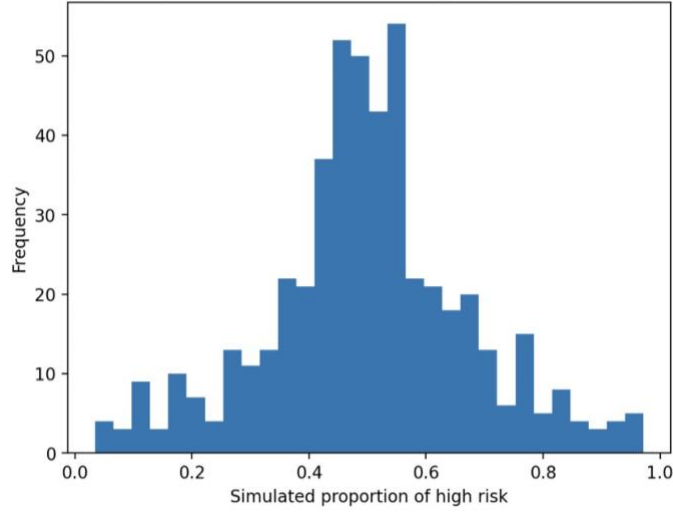


Figure 4: Prior predictive distribution of simulated high-risk prevalence

The model was fit using PyMC. We ran 4 independent chains. Each chain contained 2,000 posterior draws after 1,000 tuning steps, resulting in a total of 8,000 posterior samples. Convergence was assessed using $\hat{R}$ and bulk effective sample size (ESS). Full results are provided in the Results section.

Using the posterior samples, we simulated three intervention scenarios. The first scenario increased walking days by one standard deviation. The second scenario reduced sedentary time by one standard deviation. The third scenario evaluated a combined intervention setting specific target thresholds for both behaviors. For each scenario, predicted probabilities were calculated under the observed and modified values. These probabilities were then averaged across participants to estimate population prevalence. The difference between factual and counterfactual prevalence was defined as the absolute risk reduction (ARR). Uncertainty was summarized using 95% credible intervals.

4. Result

4.1. Frequentist Reference Model

The reference model converged successfully with a log-likelihood of -858.93 and a pseudo- R^2 of 0.079. The overall explanatory power was modest. Full results are presented in Table 1.

Table 1: Frequentist Logistic Regression Results

Variable	β	95% CI	p-value
Intercept	-0.682	[-1.022, -0.341]	<0.001
Age (centered)	0.049	[0.041, 0.058]	<0.001
Female	-0.138	[-0.392, 0.115]	0.284
Sedentary hours (centered)	0.041	[0.003, 0.079]	0.037
Vigorous activity (any)	0.107	[-0.169, 0.383]	0.447
Moderate activity days	-0.000	[-0.039, 0.039]	0.994
Walking days	-0.045	[-0.086, -0.005]	0.029
Sleep ≤ 5 h	-0.038	[-0.451, 0.376]	0.858
Sleep 6h	0.137	[-0.202, 0.476]	0.429
Sleep 8h	0.029	[-0.278, 0.336]	0.853
Sleep ≥ 9 h	0.039	[-0.372, 0.451]	0.852

At the 5% level, age, sedentary time, and walking days were statistically significant. The remaining variables did not reach statistical significance, such as sex, vigorous activity, moderate activity, and all sleep duration categories.

4.2. Bayesian Posterior Inference

We assessed MCMC convergence using trace plots and standard diagnostic statistics. As shown in Figure 5, all four chains for the three key parameters overlapped well and showed no systematic trends. This indicates good convergence. All parameters achieved $\hat{R} = 1.0$. Furthermore, bulk ESS values were well above 4,000. These metrics suggest that the posterior samples were stable and sufficient for inference.

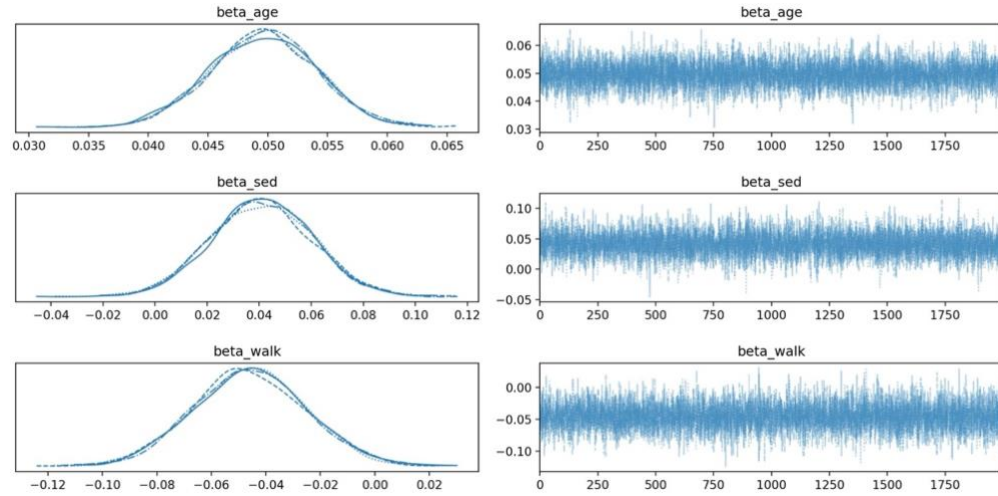


Figure 5: Trace plots

The posterior predictive check in Figure 6 shows that datasets simulated from the posterior predictive distribution closely matched the observed outcome distribution. This indicates a satisfactory overall fit. Figure 7 presents the 95% highest density intervals (HDIs) for all slope

parameters. The first three parameters had HDIs that exclude zero, suggesting credible associations with high diabetes risk.

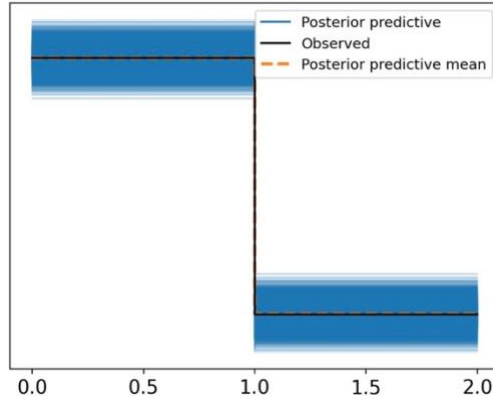


Figure 6: Posterior predictive check

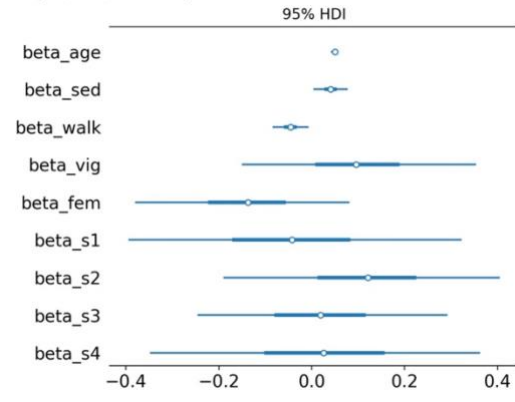


Figure 7: Forest plot

Age showed a positive association with risk, which indicates that older participants were more likely to be classified as high risk. Sedentary time was also positively associated with risk, which means that daily sitting time above the sample mean was associated with a higher probability of high diabetes risk. In contrast, walking days showed a negative association, which suggests that more frequent walking was associated with lower diabetes risk. Figure 8 shows the well concentrated and unimodal posterior densities for these three parameters.

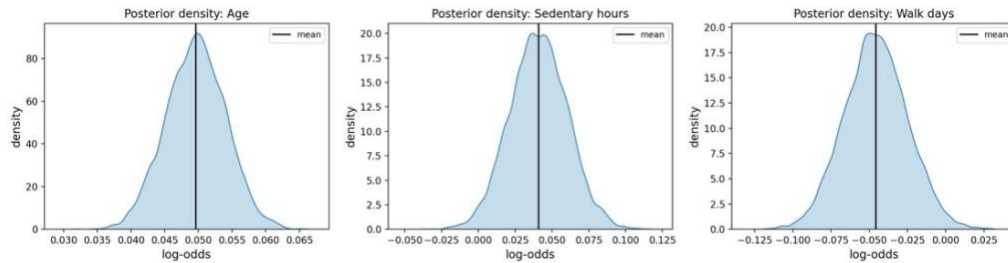


Figure 8: posterior density

The HDI of the remaining parameters such as sex, vigorous activity, and all sleep duration categories is wide and largely across zero, indicating insufficient evidence of independent associations after accounting for the other predictors. Overall, the Bayesian estimates were very similar to the frequentist results. This alignment is consistent with the weakly informative priors and the relatively large sample size.

4.3. Counterfactual Simulation

4.3.1. Counterfactual analysis on individual factors

Based on our model, the current baseline prevalence of high diabetes risk in this population is 31.2% (95% *CrI*: [29.1%, 33.3%]). This serves as the benchmark against which we measure our simulated lifestyle changes.

To accurately compare the relative impact of individual predictors, we simulated an equivalent, standardized marginal shift in behavior across the population. Under these conditions, reducing sedentary time yielded a stronger and more confident absolute risk reduction (2.3 percentage points, 95% *CrI*: [0.2,4.3]) compared to increasing walking frequency (1.5 percentage points, 95% *CrI*: [0.1,2.8]). This indicates that, on a marginal basis, minimizing prolonged sitting is a more critical intervention point for reducing population-level diabetes risk than merely increasing days of walking.

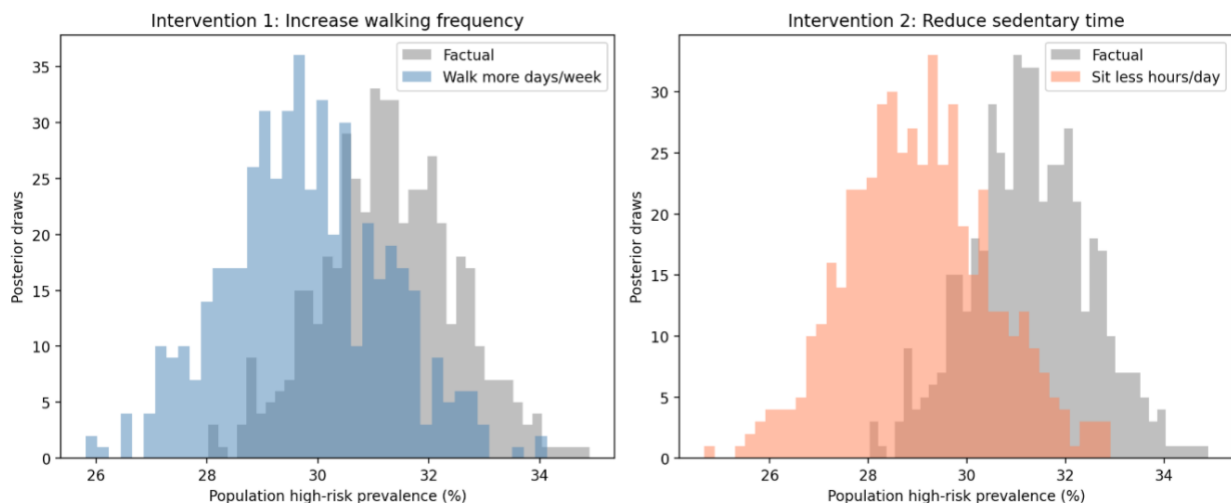


Figure 9: Counterfactual histograms

4.3.2. Counterfactual analysis on a combined intervention

To provide actionable insights for public policy, we also simulated a practical, targeted campaign: ensuring everyone walks at least 4 days a week and sits for no more than 6 hours a day.

Establishing these “minimum healthy habits” projects a distinct and definitive downward shift in the overall population risk profile. This combined intervention yields a median absolute risk reduction of 1.7 percentage points. Importantly, our draw-by-draw posterior analysis confirms this benefit with high statistical confidence (95% *CrI*: [0.6,2.9]). This demonstrates that simultaneously establishing a minimum floor for physical activity and a maximum ceiling for sedentary time provides a highly reliable, compounded improvement to overall population diabetes risk.

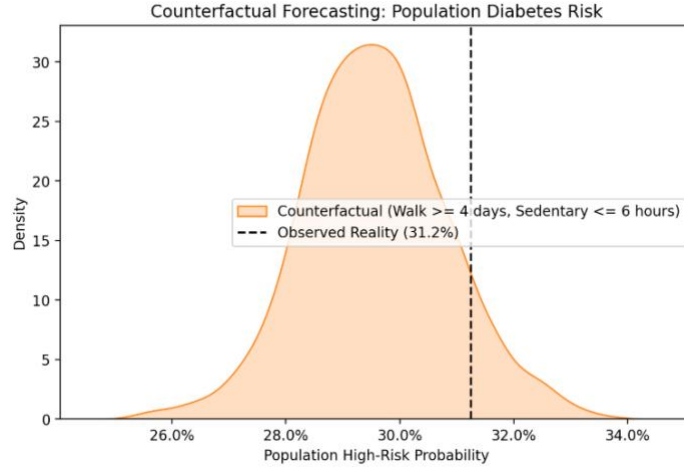


Figure 10: Combined intervention KDE

5. Discussion & Conclusion

5.1. Conclusion

To answer the first question regarding the primary drivers of high diabetes risk, both the frequentist and Bayesian logistic regression models reached the same conclusion: age, sedentary duration, and days of walking are the primary drivers. The first two exacerbate diabetes risk, while the latter demonstrates a protective effect.

Because age is a non-modifiable risk factor, we aimed to quantify the magnitude of effect for modifying sedentary behavior and walking frequency. By decreasing sedentary time by one standard deviation, increasing walking frequency by one standard deviation, and holding all other covariates at their observed values, we quantified the absolute risk reduction via a draw-to-draw comparison of the posterior predictive distributions. The isolated interventions yielded reductions of:

- 2.3 percentage points (95% *CrI*: [0.2, 4.3]) for reduced sedentary behavior.
- 1.5 percentage points (95% *CrI*: [0.1, 2.8]) for increased walking frequency.

This indicates that reducing sedentary time has a slightly stronger population-level effect. Finally, to evaluate an actionable public health campaign, we simulated a combined threshold intervention. By simulating a scenario where the general public is encouraged to walk more than 4 days a week and sit less than 6 hours a day, the population-level diabetes risk was reduced by 1.7 percentage points (95% *CrI*: [0.6, 2.9]).

5.2. Limitations and potential improvements

The primary limitation of the current mathematical framework is the use of a simplified Bayesian Logistic Regression that models age, sex, physical activity, sedentary behavior, and sleep duration as strictly independent predictors. These lifestyle factors are highly interdependent. For

instance, age fundamentally influences baseline activity levels, and increased physical activity naturally correlates with reduced sedentary time and altered sleep duration.

Future iterations of this model should incorporate interaction terms (e.g., physical activity*sedentary time) and hierarchical structures. This would allow the model to capture the compounding effects of these variables and better account for internal collinearity.

Secondly, the model's reliance on the 2024 Mexico ENSANUT survey introduces data-level limitations. Because the data is self-reported, it is inherently subject to recall bias. Furthermore, the dataset measures physical activity strictly by frequency (e.g., days of walking per week) without capturing the intensity or duration of those sessions, treating short walks and intense walks equally.

To isolate the true metabolic impact of physical activity, future studies should replace survey data with objective, longitudinal measurements collected via wearable tracking devices. This would provide precise metrics on duration, intensity (e.g., metabolic equivalents), and chronological behavioral shifts.

Finally, the cross-sectional design of the survey provides only a single snapshot in time, making it difficult to definitively establish temporal causality between lifestyle changes and diabetes risk. Critically, the dataset lacks comprehensive individual nutritional data. Because diet heavily influences both energy expenditure (activity levels) and diabetes risk, it acts as a massive unmeasured confounder.

Without nutritional data, a crucial backdoor path in our causal DAG remains open. Securing comprehensive dietary records in future datasets is essential to successfully close this path and isolate the true, unconfounded causal effect of physical activity and sedentary behavior on population health.

6. References

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[2] Hamilton, M. T., Hamilton, D. G., & Zderic, T. W. (2014). Sedentary behavior as a mediator of type 2 diabetes. *Medicine and sport science*, 60, 11–26. <https://doi.org/10.1159/000357332>

7. Appendix

Notebook: <https://github.com/hhhaiying/diabetes-lifestyle-analysis/blob/main/notebook/509final.ipynb>